

- 128.** Let the number of wickets taken till the last match be x .

Then,

$$\frac{12.4x + 26}{x + 5} = 12 \Rightarrow 12.4x + 26 = 12x + 60$$

$$\Rightarrow 0.4x = 34 \Rightarrow x = \frac{34}{0.4} = \frac{340}{4} = 85.$$

- 129.** Let the total score be x .

$$\therefore \frac{x + 92 - 85}{8} = 84 \Rightarrow x + 7 = 672 \Rightarrow x = 665.$$

- 130.** Average speed = $\frac{2xy}{x+y} = \left(\frac{2 \times 50 \times 30}{50 + 30} \right)$ km/hr = 37.5 km/hr.

- 131.** Let A , B , C , D and E represent their respective weights. Then,

$$A + B + C = (84 \times 3) = 252 \text{ kg}, A + B + C + D = (80 \times 4) = 320 \text{ kg}.$$

$$\therefore D = (320 - 252) \text{ kg} = 68 \text{ kg}, E = (68 + 3) \text{ kg} = 71 \text{ kg}.$$

$$B + C + D + E = (79 \times 4) = 316 \text{ kg}.$$

$$\text{Now, } (A + B + C + D) - (B + C + D + E) = (320 - 316) \text{ kg} = 4 \text{ kg}.$$

$$\therefore A - E = 4 \Rightarrow A = (4 + E) = 75 \text{ kg}.$$

- 132.** Sum of the present ages of husband, wife and child
 $= (23 \times 2 + 5 \times 2) + 1 = 57$ years.

$$\therefore \text{Required average} = \left(\frac{57}{3} \right) = 19 \text{ years}.$$

- 133.** Sum of the present ages of A and B
 $= (18 \times 2 + 4 \times 2) \text{ years} = 44 \text{ years}.$

$$\text{Sum of the present ages of } A, B \text{ and } C \\ = (24 \times 3) \text{ years} = 72 \text{ years}.$$

$$C's \text{ present age} = (72 - 44) \text{ years} = 28 \text{ years}.$$

$$\therefore C's \text{ age after 8 years} = (28 + 8) \text{ years} = 36 \text{ years}.$$

- 134.** Sum of the present ages of A , B , C and D
 $= (45 \times 4 + 5 \times 4) \text{ years} = 200 \text{ years}.$

$$\text{Sum of the present ages of } A, B, C, D \text{ and } X \\ = (49 \times 5) \text{ years} = 245 \text{ years}.$$

$$\therefore X's \text{ present age} = (245 - 200) \text{ years} = 45 \text{ years}.$$

- 135.** Sum of the present ages of husband, wife and child
 $= (27 \times 3 + 3 \times 3) \text{ years} = 90 \text{ years}.$

$$\text{Sum of the present ages of wife and child} \\ = (20 \times 2 + 5 \times 2) \text{ years} = 50 \text{ years}.$$

$$\therefore \text{Husband's present age} = (90 - 50) \text{ years} = 40 \text{ years}.$$

- 136.** Sum of the ages of husband and wife at the time of their marriage = $(25 \times 2) \text{ yrs} = 50 \text{ yrs}.$

$$\text{Sum of the ages of husband and wife when their son was born} = (50 + 2 \times 2) \text{ yrs} = 54 \text{ yrs}.$$

$$\text{Sum of the ages of husband, wife and son at present} \\ = (24 \times 3) \text{ years} = 72 \text{ years}.$$

$$\therefore \text{Age of son} = \frac{(72 - 54)}{3} = \frac{18}{3} = 6 \text{ years}.$$

Hence, the couple got married $(6 + 2) = 8$ years ago.

- 137.** Sum of the ages of father, mother and son at the time of son's marriage = $(42 \times 3) \text{ years} = 126 \text{ years}.$

$$\text{Sum of the present ages of father, mother and son} \\ = (126 + 3 \times 6) \text{ years} = 144 \text{ years}.$$

$$\text{Sum of the present ages of father, mother, son and grandson} \\ = (144 + 5) \text{ years} = 149 \text{ years}.$$

$$\text{Sum of the present ages of father, mother, son, daughter-in-law and grandson}$$

$$= (36 \times 5) \text{ years} = 180 \text{ years}.$$

$$\text{Daughter-in-law's present age}$$

$$= (180 - 149) \text{ years} = 31 \text{ years}.$$

$$\therefore \text{Age of daughter-in-law at the time of marriage} \\ = (31 - 6) \text{ years} = 25 \text{ years}.$$

- 138.** Sum of the ages of 4 members, 4 years ago
 $= (18 \times 4) \text{ years} = 72 \text{ years}.$

$$\text{Sum of the ages of 4 members now} \\ = (72 + 4 \times 4) \text{ years} = 88 \text{ years}.$$

$$\text{Sum of the ages of 5 members now} \\ = (18 \times 5) \text{ years} = 90 \text{ years}.$$

$$\therefore \text{Age of the baby} = (90 - 88) \text{ years} = 2 \text{ years}.$$

- 139.** Age decreased = $(5 \times 3) \text{ years} = 15 \text{ years}.$
 So, the required difference = 15 years.

- 140.** Since the total or average age of all the family members is not given, the given data is inadequate. So, the age of second child cannot be determined.

- 141.** Let the initial number of persons be x . Then,
 $16x + 20 \times 15 = 15.5(x + 20) \Leftrightarrow 0.5x = 10 \Leftrightarrow x = 20.$

- 142.** Sum of the ages of all 8 members, 10 years ago
 $= 231 \text{ years}.$

$$\text{Sum of the ages of all members, 7 years ago} \\ = (231 + 8 \times 3 - 60) \text{ years} = 195 \text{ years}.$$

$$\text{Sum of the ages of all members, 4 years ago} \\ = (195 + 8 \times 3 - 60) \text{ years} = 159 \text{ years}.$$

$$\text{Sum of the present ages of all 8 members} \\ = (159 + 8 \times 4) \text{ years} = 191 \text{ years}.$$

$$\therefore \text{Current average age} = \left(\frac{191}{8} \right) \text{ years} \\ = 23.8 \text{ years} \approx 24 \text{ years}.$$

- 143.** When the first child was born, the total age of all the family members = $(16 \times 3) \text{ years} = 48 \text{ years}.$

$$\text{When the second child was born, the total age of all the family members} = (15 \times 4) \text{ years} = 60 \text{ years}.$$

By the time the second child was born, each one of the 3 family members had grown by

$$\left(\frac{60 - 48}{3} \right) = \frac{12}{3} = 4 \text{ years}.$$

Hence, the age of eldest son when the second child was born = 4 years.

$$\text{When the third child was born, the total age of all the family members} = (16 \times 5) \text{ years} = 80 \text{ years}.$$

By the time, the third child was born, each one of the four family members had grown by $\left(\frac{80 - 60}{4} \right) = 5 \text{ years}.$